Water Footprint of India's E20 Transition: Impacts on Groundwater and Pathways for Sustainable Ethanol Production

Team NeerNexus (CM 617, Research pathway): Armaan Valecha (23B0432), Om Anghan (23B0338), Akhil Tripathi (23B0308), Vaishnavi Rajput (23B0382)

1. The question

India now blends about 20% ethanol into petrol (E20), up from 1.5% in 2013-14, and aims for 30% by 2030. The ethanol is made from sugarcane, rice and maize, all water-intensive crops, and in much of India their water is pumped from the ground. A common claim is that E20 is pushing farmers toward these crops and draining groundwater. A second force pushes the same crops: the minimum support price (MSP), the price at which the government promises to buy a crop. We asked whether E20 is raising groundwater use, and whether ethanol or MSP is the stronger driver.

2. How we went about it

We used official data for ten major sugarcane, rice and maize states: crop area and production by season (Ministry of Agriculture, 2016-17 to 2025-26), groundwater assessments of the Central Ground Water Board (2017–2024), monsoon rainfall, MSP, and ethanol volumes by feedstock. We worked in five steps, each building on the last: (1) how much of each crop goes into ethanol; (2) which crops grew after the 2018 ethanol push, and in which season; (3) whether that growth follows ethanol demand or MSP rises; (4) whether groundwater stress, the water pumped in a year as a share of what is replenished, rose where the crops expanded, allowing for rainfall; and (5) how much extra groundwater is involved and what it costs. We had planned a yearly groundwater series for 2015–2025, but assessments exist only for 2017, 2020 and 2022–2024, so we compared 2017 with 2024.

3. What we found

Step 1: maize, not sugarcane. About a fifth of India's maize went into ethanol in 2023-24 and nearly a third in 2024-25, while the shares of rice (2–4%) and sugarcane (at most 10%) stayed small. India also became a net importer of maize. So we followed maize.

Why maize: a chain of decisions. In 2022 the government brought the E20 deadline forward from 2030 to 2025-26. In 2023, to protect food supplies and prices, it stopped selling government rice to distilleries and restricted sugarcane juice for ethanol. Maize was the feedstock left open: its share of ethanol rose from almost nothing in 2021-22 to about 37% in 2025-26. No law asked farmers to grow maize; a chain of reasonable decisions did.

Step 2: maize moved into the dry season. Maize output rose 45% between 2016-19 and 2022-25. The share grown in the dry rabi and summer seasons, which depend on irrigation, rose from 27% to 43% (Figure 1a), and dry-season maize area rose 70%. With farm electricity heavily subsidised, pumping costs the farmer very little.

Step 3: MSP alone does not explain it. Millets received the largest MSP increases (106–160% from 2015-16 to 2024-25), yet their output grew only 8–39%. Maize received a smaller increase (68%), but its output grew 92%, in step with distillery demand. Rice growth matches MSP backed by government procurement, not ethanol.

Step 4: the expected damage did not show up, yet. In most states where these crops expanded, groundwater stress fell between 2017 and 2024 (Figure 1b), because the 2019–2022 monsoons were well above normal. Telangana expanded these crops the most, yet its stress fell the most, as its replenishable groundwater grew by 49%. Only Rajasthan and Madhya Pradesh worsened. With ten states no correlation was significant, so state averages could not isolate ethanol's effect; that is not the same as no effect.

Step 5: measuring the demand directly. The extra dry-season maize (12.7 lakh hectares, about 8.5 times the area of Delhi) × 300–700 mm of irrigation × 61–100% drawn from wells gives 2.3–8.9 billion m^3 of extra groundwater a year. That is only 1–4% of India's annual extraction, which is why national figures cannot see it. But in Bihar, Karnataka, Uttar Pradesh and Telangana, even the low estimate equals about 30–75% of each state's rise in extraction

since 2017. Per litre, ethanol from dry-season maize carries 860–4,070 litres of groundwater, against 3–5 litres used inside the distillery, yet groundwater rules regulate the factory while farmers are exempt. At Punjab's farm power subsidy, the pumping costs the state ₹2–10 per litre of ethanol, a cost missing from the ethanol price, while each litre saves about ₹41 of oil imports. The savings are national; the water bill is local.

Table 1. The five steps: what the claim expects and what we found

Step	What the claim expects	What we found
1. Ethanol share	Large for sugarcane	Only maize: 20% to 30%; rice 2–4%; cane ≤10%
2. Crop growth	Cane and rice expand	Maize +45%; dry-season share 27% to 43%
3. Ethanol or MSP	MSP drives crops	Ethanol fits maize; MSP plus buying fits rice
4. Groundwater	Stress rises	Fell in most; rose in Rajasthan, Madhya Pradesh
5. Extra water	Not estimated	2.3–8.9 billion m³/yr; 30–75% of rise in 4 states

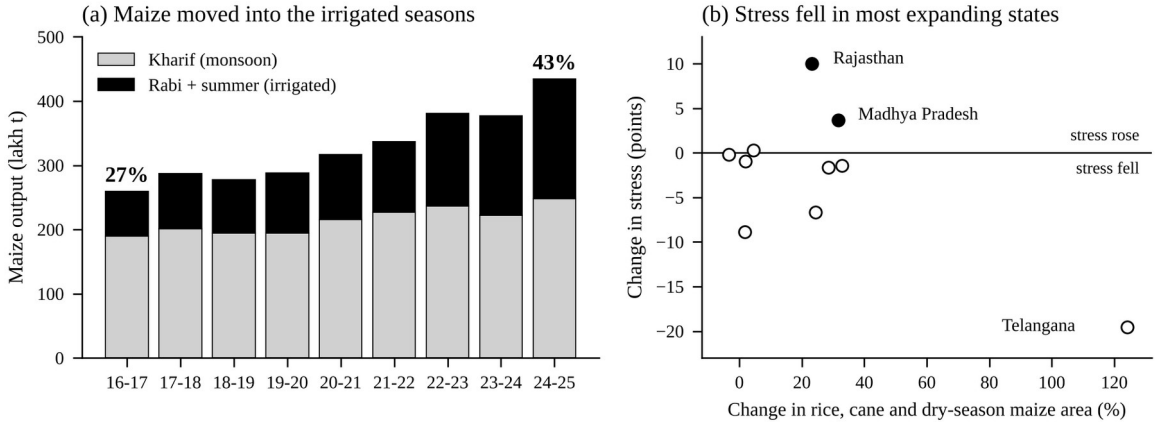


Figure 1. (a) All-India maize output by season (lakh tonnes): the irrigated rabi and summer share rose from 27% to 43%. (b) Ten states: change in crop area (2016-19 to 2021-24) against change in groundwater stress (2017 to 2024); filled points are where stress rose. Sources: UPAg; CGWB via RBI.

4. What it means

The claim is partly true. E20 is creating new groundwater demand, but through dry-season maize rather than sugarcane; so far it is small nationally and masked by good monsoons, and with blending heading to 30% the same path would enlarge the hidden share. Ethanol and MSP act more as partners than rivals: ethanol demand drives maize, while MSP works only with assured procurement, as for rice. The same chain of decisions can be tuned:

- **Pay for ethanol by season:** better prices for monsoon maize, damaged grain and crop residues, and no premium for dry-season maize from water-stressed areas.
- **MSP with a buyer:** pair MSP for millets with assured government buying in water-stressed states.
- **Track the field, not just the factory:** require distilleries to disclose where their grain comes from, so the effect can be followed district by district.

5. Limitations

Ten states and state averages; groundwater assessed in only five years; some ethanol-by-crop figures from news and industry reports; a gross water estimate that does not subtract the crops maize replaced; and a power-subsidy cost that is only a minimum. The detailed report and dataset, with every source and calculation, are at: [Drive link].

Key sources: Ministry of Agriculture, UPAg state crop data; CGWB groundwater assessments via RBI Handbook Table 107; DES MSP statement, Dec 2025; PIB on advancing E20, May 2022; Rajya Sabha reply on maize ethanol, Jul 2026.